Topic 10: Multiplication by t

Unit IV — Laplace Transforms

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Contents

1 Opening Hook

Hook

You know $\mathcal{L}\{\sin(at)\} = a/(s^2+a^2)$. But what is $\mathcal{L}\{t\sin(at)\}$? You cannot use the standard table directly — $t\sin(at)$ is not in it. You could try integrating by parts, but that is very messy. There is a clean shortcut: multiplying by t in the time domain is equivalent to differentiating $F(s)$ with respect to s (and multiplying by -1). So

$$\mathcal{L}\{t\sin(at)\} = -\frac{d}{ds}\left[\frac{a}{s^2+a^2}\right] = \frac{2as}{(s^2+a^2)^2}.$$

No integration needed. This topic proves why, and gives you a powerful tool for new transforms.

2 Why This Topic Exists

"Before we begin, here is the honest answer to why you are reading this..."

Many engineering signals involve t as a factor: damped sinusoids $te^{-at}\sin(\omega t)$, growing ramps t^2, resonance phenomena $t\cos(\omega t)$. Without the multiplication-by-t property you would have to perform tedious integration by parts every single time such a signal appears. This property converts that integration problem into a simple calculus problem—differentiate $F(s)$ and flip the sign.

Mistake	Consequence
Not knowing this property	Spending minutes on messy integration by parts for every $tf(t)$ function
Forgetting the $(-1)^n$ factor	Getting wrong signs in the result; the answer is correct in magnitude but wrong in sign for odd n
Differentiating with respect to t instead of s	Completely wrong approach — must differentiate $F(s)$ with respect to s, not $f(t)$ with respect to t

Key Takeaway

The multiplication-by-t property is a "free pass" that converts time-domain multiplication into s-domain differentiation. Master the sign convention $\mathcal{L}\{tf(t)\} = -F'(s)$ and you will never need integration by parts for this class of transforms.

3 You Already Know This (Intuition First)

Recall from calculus that differentiating e^{at} with respect to the parameter a gives te^{at}. The Laplace transform contains exactly the same mechanism. Write

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt.$$

The integrand contains e^{-st}. When you differentiate $F(s)$ with respect to s, the chain rule pulls down a factor of $(-t)$ from the exponent:

$$\frac{d}{ds}\left[e^{-st}\right] = -t\,e^{-st}.$$

So differentiating $F(s)$ once inserts one factor of $-t$ inside the integral. Repeating n times inserts $(-t)^n = (-1)^n t^n$. The entire formula is a consequence of "differentiation under the integral sign" — a classical Leibniz trick. No new theory is needed; it is the same calculus you already know, applied to a parameter inside an integral.

4 The Main Theorem — Multiplication by t

Theorem: Multiplication by t

Theorem. If $\mathcal{L}\{f(t)\} = F(s)$ for $s > \sigma_0$, then

$$\mathcal{L}\{t\,f(t)\} = -\frac{d}{ds}F(s) = -F'(s).$$

Proof (differentiation under the integral sign).
Start from the definition:

$$F(s) = \int_0^\infty e^{-st} f(t)\,dt.$$

Differentiate both sides with respect to s. Under the assumption that $f(t)$ satisfies the conditions for uniform convergence (which hold whenever f is piecewise continuous and of exponential order), the derivative may be moved inside the integral:

$$\frac{dF}{ds} = \int_0^\infty \frac{\partial}{\partial s}\left[e^{-st}\right] f(t)\,dt = \int_0^\infty (-t)\,e^{-st} f(t)\,dt = -\int_0^\infty e^{-st}\left[t\,f(t)\right]dt = -\mathcal{L}\{t\,f(t)\}.$$

Rearranging:

$$\boxed{\mathcal{L}\{t\,f(t)\} = -F'(s).}$$

5 General Formula — Multiplication by t^n

General Formula

Theorem. For any non-negative integer n,

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}F(s).$$

Derivation. Apply the $n = 1$ result repeatedly. After one application: $\mathcal{L}\{t\,f(t)\} = -F'(s)$. Apply it again with $g(t) = t\,f(t)$, so $G(s) = -F'(s)$:

$$\mathcal{L}\{t^2 f(t)\} = -\frac{d}{ds}G(s) = -\frac{d}{ds}\left[-F'(s)\right] = F''(s) = (-1)^2 F''(s).$$

Matches the formula for $n = 2$. ✓ Applying n times in this fashion gives the general result.

The $n = 2$ case explicitly: With $f(t) = e^{at}$, $F(s) = 1/(s-a)$:

$$F'(s) = -\frac{1}{(s-a)^2}, \qquad F''(s) = \frac{2}{(s-a)^3},$$

so $\mathcal{L}\{t^2 e^{at}\} = (-1)^2 F''(s) = \dfrac{2}{(s-a)^3}$.

n	**Formula**	**Example result (with $f(t) = e^{at}$)**
0	$\mathcal{L}\{f(t)\} = F(s)$	$1/(s-a)$
1	$\mathcal{L}\{tf(t)\} = -F'(s)$	$1/(s-a)^2$
2	$\mathcal{L}\{t^2 f(t)\} = F''(s)$	$2/(s-a)^3$
3	$\mathcal{L}\{t^3 f(t)\} = -F'''(s)$	$6/(s-a)^4$
4	$\mathcal{L}\{t^4 f(t)\} = F^{(4)}(s)$	$24/(s-a)^5$

6 Key Application Transforms

Deriving Important Transforms via Multiplication by t

1. $\mathcal{L}\{te^{at}\}$.
Here $f(t) = e^{at}$, so $F(s) = 1/(s-a)$. Differentiate:

$$F'(s) = \frac{d}{ds}\frac{1}{s-a} = -\frac{1}{(s-a)^2}.$$

Therefore

$$\mathcal{L}\{te^{at}\} = -F'(s) = \frac{1}{(s-a)^2}.$$

2. $\mathcal{L}\{t\sin(at)\}$.
Here $f(t) = \sin(at)$, $F(s) = a/(s^2+a^2)$. Apply the quotient rule:

$$F'(s) = \frac{d}{ds}\frac{a}{s^2+a^2} = \frac{0\cdot(s^2+a^2) - a\cdot 2s}{(s^2+a^2)^2} = \frac{-2as}{(s^2+a^2)^2}.$$

Therefore

$$\mathcal{L}\{t\sin(at)\} = -F'(s) = \frac{2as}{(s^2+a^2)^2}.$$

3. $\mathcal{L}\{t\cos(at)\}$.
Here $F(s) = s/(s^2+a^2)$. Quotient rule:

$$F'(s) = \frac{1\cdot(s^2+a^2) - s\cdot 2s}{(s^2+a^2)^2} = \frac{s^2+a^2-2s^2}{(s^2+a^2)^2} = \frac{a^2-s^2}{(s^2+a^2)^2}.$$

Therefore

$$\mathcal{L}\{t\cos(at)\} = -F'(s) = \frac{s^2-a^2}{(s^2+a^2)^2}.$$

4. $\mathcal{L}\{t^2 e^{at}\}$.
$F(s) = 1/(s-a)$, $F'(s) = -1/(s-a)^2$, $F''(s) = 2/(s-a)^3$. Thus

$$\mathcal{L}\{t^2 e^{at}\} = (-1)^2 F''(s) = \frac{2}{(s-a)^3}.$$

5. $\mathcal{L}\{t^n e^{at}\}$ — General result.
Since $F^{(n)}(s) = (-1)^n\, n!/(s-a)^{n+1} \cdot (-1)^0$, applying $(-1)^n d^n/ds^n$ to $F(s) = 1/(s-a)$:

$$F^{(k)}(s) = \frac{(-1)^k k!}{(s-a)^{k+1}},$$

so

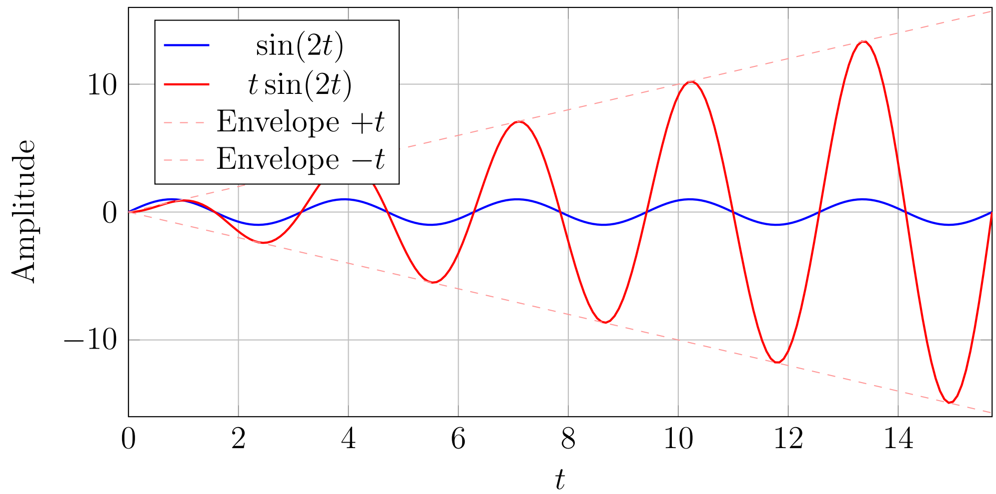
$$\mathcal{L}\{t^n e^{at}\} = (-1)^n F^{(n)}(s) = (-1)^n \cdot \frac{(-1)^n n!}{(s-a)^{n+1}} = \frac{n!}{(s-a)^{n+1}}.$$

Verification: $n=1$: $1!/(s-a)^2 = 1/(s-a)^2$ ✓; $n=2$: $2!/(s-a)^3 = 2/(s-a)^3$ ✓; $n=3$: $3!/(s-a)^4 = 6/(s-a)^4$ ✓.

7 Visualizing the Property

The graph below shows how multiplication by t converts a bounded sinusoid into a signal with linearly growing amplitude — a classic resonance signature.

$f(t) = \sin(2t)$ versus $g(t) = t\sin(2t)$ for $t \in [0, 5\pi]$



Caption: $t\sin(2t)$ represents a resonance phenomenon — amplitude grows linearly with time. The factor t in the time domain corresponds to differentiation in the s-domain.

8 Worked Examples

Example 1: Find $\mathcal{L}\{t^2\cos(3t)\}$

We use $\mathcal{L}\{t^2 f(t)\} = F''(s)$ with $f(t) = \cos(3t)$, $F(s) = s/(s^2+9)$.
Step 1 — First derivative (quotient rule with $u = s$, $v = s^2+9$):

$$F'(s) = \frac{1\cdot(s^2+9) - s\cdot 2s}{(s^2+9)^2} = \frac{9-s^2}{(s^2+9)^2}.$$

Step 2 — Second derivative (quotient rule again with $u = 9-s^2$, $v = (s^2+9)^2$):

$$\frac{d}{ds}(9-s^2) = -2s, \qquad \frac{d}{ds}(s^2+9)^2 = 2(s^2+9)\cdot 2s = 4s(s^2+9).$$

$$F''(s) = \frac{-2s\,(s^2+9)^2 - (9-s^2)\cdot 4s(s^2+9)}{(s^2+9)^4} = \frac{-2s(s^2+9) - 4s(9-s^2)}{(s^2+9)^3}.$$

Expand the numerator:

$$-2s^3 - 18s - 36s + 4s^3 = 2s^3 - 54s = 2s(s^2-27).$$

Therefore

$$\boxed{\mathcal{L}\{t^2\cos(3t)\} = F''(s) = \frac{2s(s^2-27)}{(s^2+9)^3}.}$$

What Did We Learn? (Example 1)

For $t^2 f(t)$ we apply two derivatives of $F(s)$ (with the net sign $(-1)^2 = +1$). The quotient rule must be applied carefully twice; keeping the intermediate expressions factored avoids arithmetic errors.

Example 2: Find $\mathcal{L}\{te^{-2t}\sin(t)\}$

Two transforms are combined here: the First Shifting Theorem and multiplication by t.
Step 1 — Identify the base function. Let $g(t) = t\sin(t)$. We want $\mathcal{L}\{e^{-2t}g(t)\}$. By the First Shifting Theorem, if $\mathcal{L}\{g(t)\} = G(s)$, then $\mathcal{L}\{e^{-2t}g(t)\} = G(s+2)$.
Step 2 — Compute $G(s) = \mathcal{L}\{t\sin(t)\}$. With $f(t) = \sin(t)$, $F(s) = 1/(s^2+1)$:

$$F'(s) = \frac{-2s}{(s^2+1)^2}, \qquad G(s) = -F'(s) = \frac{2s}{(s^2+1)^2}.$$

Step 3 — Apply the shift $s \to s+2$.

$$\mathcal{L}\{te^{-2t}\sin(t)\} = G(s+2) = \frac{2(s+2)}{\left((s+2)^2+1\right)^2} = \frac{2(s+2)}{(s^2+4s+5)^2}.$$

Therefore

$$\boxed{\mathcal{L}\{te^{-2t}\sin(t)\} = \frac{2(s+2)}{(s^2+4s+5)^2}.}$$

What Did We Learn? (Example 2)

When e^{at} and a t-factor both appear, first use multiplication by t to find $G(s)$, then shift $s \to s - a$ (here $s \to s + 2$). The two properties compose neatly.

Example 3: Given $\mathcal{L}\{f(t)\} = \ln(s^2 + a^2)$, find $\mathcal{L}\{tf(t)\}$

We use $\mathcal{L}\{tf(t)\} = -F'(s)$ with $F(s) = \ln(s^2 + a^2)$.
Differentiate $F(s)$:

$$F'(s) = \frac{d}{ds} \ln(s^2 + a^2) = \frac{2s}{s^2 + a^2}.$$

Apply the formula:

$$\mathcal{L}\{tf(t)\} = -F'(s) = -\frac{2s}{s^2 + a^2}.$$

Pedagogical Note: In rigorous transform theory, a valid Laplace transform of an exponential-order function must satisfy $\lim_{s\to\infty} F(s) = 0$. Because $\lim_{s\to\infty} \ln(s^2 + a^2) = \infty$, no classical function possesses $\ln(s^2 + a^2)$ as its standard Laplace transform. However, this problem is a standard formal operational exercise in transform calculus to practice applying the differentiation rule $\mathcal{L}\{tf(t)\} = -F'(s)$ algebraically.

What Did We Learn? (Example 3)

The multiplication-by-t formula works purely algebraically on any differentiable $F(s)$ — only $F'(s)$ needs to be computed. Here $F(s) = \ln(s^2 + a^2)$ differentiates easily by the chain rule to give $-2s/(s^2 + a^2)$.

9 Extended Standard Table

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
t	$\dfrac{1}{s^2}$
t^2	$\dfrac{2}{s^3}$
t^n	$\dfrac{n!}{s^{n+1}}$
te^{at}	$\dfrac{1}{(s-a)^2}$
$t^n e^{at}$	$\dfrac{n!}{(s-a)^{n+1}}$
$t\sin(at)$	$\dfrac{2as}{(s^2+a^2)^2}$
$t\cos(at)$	$\dfrac{s^2-a^2}{(s^2+a^2)^2}$
$t\sinh(at)$	$\dfrac{2as}{(s^2-a^2)^2}$
$t\cosh(at)$	$\dfrac{s^2+a^2}{(s^2-a^2)^2}$

(All entries for $t\cosh$ and $t\sinh$ are derived using $\mathcal{L}\{\sinh(at)\} = a/(s^2-a^2)$ and $\mathcal{L}\{\cosh(at)\} = s/(s^2-a^2)$ via the same quotient rule.)

10 Excel Example

Numerical Verification of $\mathcal{L}\{t\sin(t)\}$ at $s = 2$

The exact value is

$$\mathcal{L}\{t\sin(t)\}\Big|_{s=2} = \frac{2s}{(s^2+1)^2}\Bigg|_{s=2} = \frac{4}{25} = 0.16.$$

We approximate the integral numerically using the trapezoid rule with step $h = 0.1$ from $t = 0$ to $t = 8$.

Column A t	**Column B** $e^{-2t}\cdot t\sin(t)$	**Column C** Cumulative Trapezoid Sum	**Remark**
0.0	`=EXP(-2*A2)*A2*SIN(A2)` = 0.000000	0.000000	Initial valu
0.1	`=EXP(-2*A3)*A3*SIN(A3)` = 0.008181	`=C2+(B2+B3)/2*0.1` = 0.000409	Trapezoid ste
0.2	`=EXP(-2*A4)*A4*SIN(A4)` = 0.026889	`=C3+(B3+B4)/2*0.1` = 0.002162	Trapezoid ste

Spreadsheet setup instructions:

1. Enter t values in Column A: cell A2 = 0, A3 = 0.1, A4 = 0.2, ..., A82 = 8.0. Use fill-down.
2. Column B formula (enter in B2, fill to B82): `=EXP(-2*A2)*A2*SIN(A2)`

3. Column C (cumulative trapezoid sum): C2 = 0, C3 = `=C2+(B2+B3)/2*0.1`, fill to C82.

4. Final approximation: value in cell C82.

Excel vs Exact Value

With $h = 0.1$ the trapezoid rule gives approximately ≈ 0.1598 at $t = 8$. The exact value is 0.16. The agreement to three significant figures confirms both the formula and the numerical method. Reducing h to 0.01 or extending to $t = 12$ brings the approximation even closer to 0.16.

11 Python Example

Listing 1: Symbolic verification of multiplication-by-t property using SymPy

```python
import sympy as sp

s, t, a, n = sp.symbols('s t a n', positive=True)

# --- Case 1: f(t) = sin(at) ---
F_sin = sp.laplace_transform(sp.sin(a*t), t, s, noconds=True)
print("L{sin(at)} =", F_sin)            # a/(s**2 + a**2)
neg_dF_sin = -sp.diff(F_sin, s)
neg_dF_sin_simplified = sp.simplify(neg_dF_sin)
print("-d/ds L{sin(at)} =", neg_dF_sin_simplified)
# Expected: 2*a*s/(s**2 + a**2)**2

# Verify against direct transform of t*sin(at)
L_tsin = sp.laplace_transform(t*sp.sin(a*t), t, s, noconds=True)
print("L{t*sin(at)} =", sp.simplify(L_tsin))
# Expected: 2*a*s/(s**2 + a**2)**2
print("Match:", sp.simplify(neg_dF_sin - L_tsin) == 0)   # True

print()

# --- Case 2: f(t) = e^(at) ---
F_exp = sp.laplace_transform(sp.exp(a*t), t, s, noconds=True)
print("L{e^(at)} =", F_exp)              # 1/(-a + s)
neg_dF_exp = sp.simplify(-sp.diff(F_exp, s))
print("-d/ds L{e^(at)} =", neg_dF_exp)
# Expected: 1/(s - a)**2

L_texp = sp.laplace_transform(t*sp.exp(a*t), t, s, noconds=True)
print("L{t*e^(at)} =", sp.simplify(L_texp))
# Expected: 1/(s - a)**2
print("Match:", sp.simplify(neg_dF_exp - L_texp) == 0)   # True

print()

# --- Case 3: f(t) = cos(at) ---
```

```python
F_cos = sp.laplace_transform(sp.cos(a*t), t, s, noconds=True)
print("L{cos(at)} =", F_cos)            # s/(s**2 + a**2)
neg_dF_cos = sp.simplify(-sp.diff(F_cos, s))
print("-d/ds L{cos(at)} =", neg_dF_cos)
# Expected: (s**2 - a**2)/(s**2 + a**2)**2

L_tcos = sp.laplace_transform(t*sp.cos(a*t), t, s, noconds=True)
print("L{t*cos(at)} =", sp.simplify(L_tcos))
# Expected: (s**2 - a**2)/(s**2 + a**2)**2
print("Match:", sp.simplify(neg_dF_cos - L_tcos) == 0)   # True

print()

# --- General formula for t^n * e^(at), n = 1, 2, 3 ---
F_e = 1/(s - a)   # L{e^(at)}
for k in [1, 2, 3]:
    Fn = ((-1)**k) * sp.diff(F_e, s, k)
    print(f"L{{t^{k} * e^(at)}} = {sp.simplify(Fn)}")
# Expected:
# L{t^1 * e^(at)} =  1/(s - a)**2
# L{t^2 * e^(at)} =  2/(s - a)**3
# L{t^3 * e^(at)} =  6/(s - a)**4
```

What Did We Learn? (Python)

SymPy's `laplace_transform` and `diff` confirm all the multiplication-by-t results symbolically. The loop over $n = 1, 2, 3$ shows the pattern $n!/(s-a)^{n+1}$ emerging automatically. Symbolic verification is a powerful sanity check before applying these results numerically.

12 Viva-Style Oral Questions

1. **State the multiplication-by-t formula.**
   Answer. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{tf(t)\} = -F'(s) = -dF/ds$.

2. **How is the formula proved?**
   Answer. Start from $F(s) = \int_0^\infty e^{-st} f(t)\,dt$. Differentiate both sides with respect to s. By the Leibniz rule (differentiation under the integral sign), the derivative passes inside the integral and $\partial/\partial s\,[e^{-st}] = -te^{-st}$. This gives $F'(s) = -\mathcal{L}\{tf(t)\}$, hence $\mathcal{L}\{tf(t)\} = -F'(s)$.

3. **Where does the $(-1)^n$ factor come from in the general formula?**
   Answer. Each differentiation with respect to s brings out one factor of $(-t)$ from e^{-st}. After n differentiations, the cumulative factor is $(-t)^n = (-1)^n t^n$. Moving $(-1)^n$ outside the integral gives the sign factor in $(-1)^n d^n F/ds^n$.

4. **What is $\mathcal{L}\{t\sin(at)\}$?**
   Answer. $\mathcal{L}\{t\sin(at)\} = 2as/(s^2 + a^2)^2$. This follows from differentiating $F(s) = a/(s^2 + a^2)$ with the quotient rule and negating.

5. **What is $\mathcal{L}\{t\cos(at)\}$?**
Answer. $\mathcal{L}\{t\cos(at)\} = (s^2 - a^2)/(s^2 + a^2)^2$. Differentiate $F(s) = s/(s^2 + a^2)$ via the quotient rule: $F'(s) = (a^2 - s^2)/(s^2 + a^2)^2$; then negate.

6. **How do you handle a function like $te^{-3t}\cos(2t)$?**
Answer. Two steps: (i) find $G(s) = \mathcal{L}\{t\cos(2t)\} = (s^2 - 4)/(s^2 + 4)^2$ using multiplication by t; (ii) apply the First Shifting Theorem with $a = -3$ to replace s by $s + 3$: $G(s + 3) = ((s + 3)^2 - 4)/((s + 3)^2 + 4)^2$.

7. **What is the physical interpretation of $t\sin(\omega t)$?**
Answer. $t\sin(\omega t)$ represents a resonance phenomenon in which the amplitude of oscillation grows linearly with time. In control systems and vibrations, this corresponds to the response of a system driven at its natural frequency ω with no damping — energy builds up indefinitely.

8. **What is the general result for $\mathcal{L}\{t^n e^{at}\}$?**
Answer. $\mathcal{L}\{t^n e^{at}\} = n!/(s - a)^{n+1}$. It follows from applying the general formula $(-1)^n d^n/ds^n$ to $F(s) = 1/(s - a)$ and computing $F^{(n)}(s) = (-1)^n n!/(s - a)^{n+1}$.

13 Descriptive Questions

1. **Prove the multiplication-by-t formula $\mathcal{L}\{tf(t)\} = -F'(s)$.**
Answer. Define $F(s) = \int_0^\infty e^{-st} f(t)\, dt$. Differentiate both sides with respect to s, invoking the Leibniz rule to move the derivative inside the integral (valid when f is of exponential order and piecewise continuous):

$$F'(s) = \int_0^\infty \frac{\partial}{\partial s}[e^{-st}]f(t)\, dt = \int_0^\infty (-t)e^{-st} f(t)\, dt = -\mathcal{L}\{tf(t)\}.$$

Rearranging gives $\mathcal{L}\{tf(t)\} = -F'(s)$. □

2. **Prove the general formula $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$.**
Answer. Proceed by induction. Base case $n = 1$ is proved above. Assume $\mathcal{L}\{t^k f(t)\} = (-1)^k F^{(k)}(s)$ holds. Let $g(t) = t^k f(t)$, $G(s) = (-1)^k F^{(k)}(s)$. By the $n = 1$ result,

$$\mathcal{L}\{t^{k+1} f(t)\} = \mathcal{L}\{t \cdot g(t)\} = -G'(s) = -(-1)^k F^{(k+1)}(s) = (-1)^{k+1} F^{(k+1)}(s).$$

This completes the induction. □

3. **Find $\mathcal{L}\{t^2\sin(at)\}$ completely.**
Answer. We need $(-1)^2 F''(s) = F''(s)$ with $F(s) = a/(s^2 + a^2)$.

First derivative (quotient rule): $F'(s) = -2as/(s^2 + a^2)^2$.

Second derivative: let $u = -2as$, $v = (s^2 + a^2)^2$. $u' = -2a$, $v' = 4s(s^2 + a^2)$.

$$F''(s) = \frac{-2a(s^2 + a^2)^2 - (-2as) \cdot 4s(s^2 + a^2)}{(s^2 + a^2)^4} = \frac{-2a(s^2 + a^2) + 8as^2}{(s^2 + a^2)^3} = \frac{2a(3s^2 - a^2)}{(s^2 + a^2)^3}.$$

Therefore $\mathcal{L}\{t^2\sin(at)\} = \dfrac{2a(3s^2 - a^2)}{(s^2 + a^2)^3}$.

4. **Find $\mathcal{L}\{te^{-t}\cos(2t)\}$.**
Answer. Step 1: find $G(s) = \mathcal{L}\{t\cos(2t)\}$. With $F(s) = s/(s^2+4)$, quotient rule gives $F'(s) = (4-s^2)/(s^2+4)^2$, so $G(s) = -F'(s) = (s^2-4)/(s^2+4)^2$.
Step 2: shift $s \to s+1$ (First Shifting Theorem with $a = -1$):
$$\mathcal{L}\{te^{-t}\cos(2t)\} = G(s+1) = \frac{(s+1)^2-4}{((s+1)^2+4)^2} = \frac{s^2+2s-3}{(s^2+2s+5)^2}.$$

5. **Derive $\mathcal{L}\{t^n\}$ using the multiplication-by-t property and $\mathcal{L}\{1\} = 1/s$.**
Answer. Let $f(t) = 1$, $F(s) = 1/s$. Apply the general formula:
$$\frac{d^n}{ds^n}\left(\frac{1}{s}\right) = \frac{(-1)^n n!}{s^{n+1}}.$$
Therefore
$$\mathcal{L}\{t^n\} = (-1)^n \cdot \frac{(-1)^n n!}{s^{n+1}} = \frac{n!}{s^{n+1}}.$$
This matches the known result for $\mathcal{L}\{t^n\}$, confirming the formula.

14 Practice Problems

1. Find $\mathcal{L}\{t\cosh(2t)\}$.
Hint: Differentiate $F(s) = s/(s^2-4)$; apply quotient rule; negate. *Answer:* $(s^2+4)/(s^2-4)^2$.

2. Find $\mathcal{L}\{t^2\sin(t)\}$.
Hint: Use $F''(s)$ with $F(s) = 1/(s^2+1)$; differentiate twice. *Answer:* $2(3s^2-1)/(s^2+1)^3$.

3. Find $\mathcal{L}\{te^{3t}\cos(t)\}$.
Hint: First compute $G(s) = \mathcal{L}\{t\cos(t)\} = (s^2-1)/(s^2+1)^2$; then shift $s \to s-3$. *Answer:* $((s-3)^2-1)/((s-3)^2+1)^2$.

4. Find $\mathcal{L}\{t^3e^{-2t}\}$.
Hint: Use $\mathcal{L}\{t^n e^{at}\} = n!/(s-a)^{n+1}$ with $n = 3$, $a = -2$. *Answer:* $6/(s+2)^4$.

5. Find $\mathcal{L}\{t\sinh(3t)\}$.
Hint: Differentiate $F(s) = 3/(s^2-9)$; negate. *Answer:* $6s/(s^2-9)^2$.

6. Given $\mathcal{L}\{f(t)\} = \arctan(1/s)$, find $\mathcal{L}\{tf(t)\}$.
Hint: Differentiate $F(s) = \arctan(1/s)$ using chain rule: $dF/ds = \frac{1}{1+(1/s)^2}\cdot(-1/s^2) = -1/(s^2+1)$. Negate. *Answer:* $1/(s^2+1)$.

15 MCQs

1. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{t^n f(t)\}$ equals
(a) $(-1)^n F(s)$ (b) $(-1)^n F^{(n)}(s)$ (c) $n!\,F(s)$ (d) $F^{(n)}(s)$
Correct answer: (b).
Explanation: Each differentiation with respect to s produces a factor of $(-t)$; after n steps the sign factor is $(-1)^n$.

2. The transform $\mathcal{L}\{t\sin(at)\}$ is

   (a) $\frac{a}{(s^2+a^2)^2}$ (b) $\frac{2as}{(s^2+a^2)^2}$ (c) $\frac{s}{s^2+a^2}$ (d) $\frac{2a}{s^2+a^2}$

   Correct answer: (b).
   Explanation: Differentiate $a/(s^2+a^2)$ and negate; quotient rule gives $-(-2as)/(s^2+a^2)^2 = 2as/(s^2+a^2)^2$.

3. $\mathcal{L}\{te^{at}\}$ equals

   (a) $\frac{1}{s-a}$ (b) $\frac{a}{(s-a)^2}$ (c) $\frac{1}{(s-a)^2}$ (d) $\frac{1}{(s+a)^2}$

   Correct answer: (c).
   Explanation: Differentiate $F(s) = 1/(s-a)$ to get $-1/(s-a)^2$; negate to obtain $1/(s-a)^2$.

4. To apply the multiplication-by-t property, you differentiate with respect to

   (a) t (b) a (c) s (d) ω

   Correct answer: (c).
   Explanation: The property is derived by differentiating $F(s)$ with respect to the frequency variable s, not with respect to t.

5. Using $\mathcal{L}\{1\} = 1/s$, the transform $\mathcal{L}\{t^2\}$ via the multiplication-by-t formula is

   (a) $\frac{1}{s^2}$ (b) $\frac{1}{s^3}$ (c) $\frac{2}{s^3}$ (d) $\frac{2}{s^2}$

   Correct answer: (c).
   Explanation: $F(s) = 1/s$; $F''(s) = 2/s^3$; $\mathcal{L}\{t^2\} = (-1)^2F''(s) = 2/s^3$.

16 Common Mistakes

Common Mistakes in Multiplication by t

Mistake	What Students Do	Correct Approach
Differentiate w.r.t. t instead of s	Compute $d/dt[f(t)]$ and use that as the transformed result	Always differentiate $F(s)$ with respect to s, never touch $f(t)$
Forget the $(-1)^n$ sign	Write $\mathcal{L}\{t^n f\} = F^{(n)}(s)$ without the sign factor; get wrong signs for odd n	Include $(-1)^n$; for $n = 1$ the answer is $-F'(s)$, not $+F'(s)$
Apply formula without n derivatives	Differentiate once even when computing $\mathcal{L}\{t^2 f\}$, stopping at $F'(s)$ instead of $F''(s)$	For $t^n f(t)$ differentiate $F(s)$ exactly n times
Sign error in quotient rule	Mix up numerator terms when differentiating rational $F(s)$; e.g. write $F'(s) = (u'v + uv')/v^2$ instead of $(u'v - uv')/v^2$	Quotient rule: $F'(s) = (u'v - uv')/v^2$; write out u, v, u', v' explicitly before substituting
Use $+F'(s)$ for $\mathcal{L}\{tf(t)\}$	Write $\mathcal{L}\{tf(t)\} = +F'(s)$, forgetting the negative sign	The formula is $\mathcal{L}\{tf(t)\} = -F'(s)$; the minus sign comes from $\partial/\partial s[e^{-st}] = -te^{-st}$

17 Quick Recap

Quick Recap — Multiplication by t

- **Main formula:** $\mathcal{L}\{tf(t)\} = -F'(s)$, where $F(s) = \mathcal{L}\{f(t)\}$.
- **General formula:** $\mathcal{L}\{t^n f(t)\} = (-1)^n \dfrac{d^n}{ds^n} F(s)$; the sign alternates with n.
- **Key results:** $\mathcal{L}\{te^{at}\} = 1/(s-a)^2$; $\mathcal{L}\{t\sin(at)\} = 2as/(s^2+a^2)^2$; $\mathcal{L}\{t\cos(at)\} = (s^2-a^2)/(s^2+a^2)^2$.
- **Resonance interpretation:** $t\sin(\omega t)$ has linearly growing amplitude — this is the mathematical signature of an undamped resonance; its transform $2\omega s/(s^2+\omega^2)^2$ encodes that growth in the s-domain.
- **Extended standard table:** Use the table in Section 8 for rapid lookup of $t\sinh$, $t\cosh$, $t^n e^{at}$, t^n without re-deriving.
- **Combined $e^{at} \cdot t \cdot f(t)$:** First apply multiplication by t to get $G(s)$, then shift $s \to s-a$ by the First Shifting Theorem.
- **Sign discipline:** Always write out u, v, u', v' in the quotient rule before substituting; this prevents the most common sign errors.

- **Verification:** Cross-check with SymPy (`sp.diff(F,s)`) or the extended table; numerical Excel integration confirms to 3–4 significant figures for reasonable step sizes.